Javed Iqbal

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Research Profile

Early-career planetary researcher with a background in exoplanetary astronomy and interests in planetary formation and evolution, planetary atmospheres, planetary habitability, and the chemical processes shaping planetary environments.

Education

- **M.S. Astronomy and Astrophysics** *2019–2023*
  Institute of Space Technology (IST), Islamabad, Pakistan *CGPA: 3.24 / 4.0*

  Thesis: *Photometric Search for Jupiter-Sized Short-Period Exoplanets in Open Clusters NGC 884 and NGC 869*
  Supervisors: Dr. Shaukat Goderya (Tarleton State University, USA); Dr. Fazeel Mahmood Khan (NYU Abu Dhabi, UAE)

  - Conducted a 30.5-hour photometric survey using a 0.8-m telescope; reduced and analyzed over 200 stellar light curves with ISIS, VaST, and custom Python scripts.
  - Identified 25 variable stars, including four new candidate variables.
  - Supported by a competitive Master's Research Grant (IST, 2022).

- **B.S. Physics** *2015–2019*
  Bahauddin Zakariya University, Multan, Pakistan *CGPA: 3.60 / 4.0*

  Final Year Project: *Comparative Analysis of Habitable Zones around M-Dwarf Systems*
  Supervisor: Dr. Tariq Masood Chaudhry

Research Experience

- **Research Collaborator** *2024–Present*
  ÇOMÜ Ulupınar Observatory, Turkey / PI: Mehmet Yüksek

  - Currently developing a Python pipeline integrating Bayesian inference (MCMC with `emcee`) and N-body simulations for transit timing variation (TTV) analysis of short-period exoplanets.
  - Analyzing timing variations of three short-period exoplanets to refine orbital ephemerides and constrain potential additional planetary companions.
  - Combining new photometric observations with archival data to build a comprehensive timing baseline.

- **Research Intern** *May–July 2026*
  Sharjah Space & Astronomy Hub (SSAH), University of Sharjah, UAE

  - Developed a modular Python-based optical data processing pipeline for Space Situational Awareness (SSA) observations using telescope FITS images.
  - Implemented image calibration, quality control, astrometric and photometric analysis, object detection, satellite tracking, and automated reporting modules.
  - Developed pipeline components for FITS header validation, Gaia DR3 astrometric calibration, timing corrections, and TLE/SGP4-based identification of known space objects.

- **Independent Research: Galactic HI 21-cm Emission** *2025*
  Self-directed; observations via PICTOR Radio Telescope (1.5-m)

  - Performed drift-scan radio observations to detect Galactic HI emission.
  - Developed Python routines for spectral calibration, RFI mitigation, Gaussian line fitting, and velocity estimation.

- Derived HI column densities consistent within ∼15% of LAB survey values; results archived on Zenodo.

- **Research Associate** *2022–2023*
  Institute of Space Technology, Islamabad, Pakistan
  - Supported observational campaigns and photometric data reduction for variable star and exoplanet transit studies using the 16-inch Cassegrain telescope.

Publications

First-Author Refereed

- **Iqbal, J.**, Goderya, S., & Mahmood Khan, F. (2024). Photometric Search for Short-Period Jupiter-Size Exoplanets in Open Clusters NGC 884 and NGC 869. *Open European Journal on Variable Stars*, 256, 1–15. [Article]

Co-authored Refereed

- Lynch, K., Aguilera, E. J., Bell, I., **Iqbal, J.**, et al. (2025). New Transit Analysis of WASP-10b. *PACS E Lab*, in press. *(Contribution: transit photometry analysis.)*

Submitted / Preprints

- **Iqbal, J.**, Masood, G., Hasnain, A. Condensation-temperature trends in FGK exoplanet hosts: no evidence for a planetary fingerprint beyond Galactic chemical evolution. *Submitted, 2026*
- **Iqbal, J.**, Goderya, S., & Sohaib, A. The Radius Valley Across FGK Dwarfs: Detection Is Not Measurement. *Submitted, 2026*
- **Iqbal, J.** Stellar Flare Activity of M-Dwarf Stars in the Pleiades Cluster from TESS Photometry. *In preparation, 2026*
- Das, B., **Iqbal, J.**, et al. Thermo-mechanical performance of a neuro-responsive regenerative window system for space habitats. Preprint. [Preprint] *(Contribution: COMSOL/ANSYS simulations.)*
- **Iqbal, J.** (2025). Detection and kinematic analysis of Galactic HI 21-cm emission using a small-aperture radio telescope. *Zenodo*. DOI: 10.5281/zenodo.17420622

Presentations and Talks

Poster Presentations

- **Photometry and Transit Modeling of the Exoplanet TOI-2109b.** [Poster] *2024*
  Sagan Exoplanet Summer Workshop, NASA Exoplanet Science Institute, Caltech
- **Determining Parameters of Open Clusters NGC 884 and NGC 869 Using Gaia EDR3.** [Poster] *2024*
  ICRAC Conference, COMSATS University Islamabad

Invited Talks

- **Life Beyond Earth: How Close Are We to Finding It?** *2025*
  Astronomy and Astrophysics Chapter, Pakistan
- **Exoplanets: Paving the Way to Search for Alien Life** *2024*
  World Space Week Seminar, NUST Islamabad
- **Search for Short-Period Jupiter-Sized Exoplanets in Open Clusters** *2023*
  World Space Week Seminar, NUST Islamabad
- **Photometric Exoplanet Detection Using TESS Data** *2023*
  IST Space Society, Islamabad

Open-Source Projects

- **Exoplanet Transit Simulator** — Interactive Tool *2023–Present*
  JavaScript-based simulator for multi-planet transit configurations and habitable-zone visualization; used in teaching and public outreach.

- **Lunar and Planetary Phase Simulator** — Interactive Tool *2023–Present*
  Browser-based simulator for reflected-light phase curves and orbital geometry of planetary systems.

Technical Skills

Proficient: Python, Bash, Git; NumPy, Pandas, SciPy, Astropy, Matplotlib; Bayesian inference, MCMC (`emcee`), Gaussian mixture models, kernel density estimation; AstroImageJ, EXOTIC, PyTransit, Batman, Allesfitter, ISIS, VaST, FITSH, VARTOOLS.

Familiar: COMSOL, ANSYS (thermo-mechanical simulation); PHOEBE (eclipsing binary modeling); radio spectral analysis (PICTOR pipeline).

Observational: CCD photometry, observation planning, telescope operation; 16-inch Cassegrain (IST) and 0.8-m Ritchey–Chrétien telescopes; drift-scan radio observations.

Professional Development

- ExoClock Unlocked Seminar Series (ExoClock Project; ESA Ariel Space Mission) *2026*
- NASA Sagan Exoplanet Summer Workshops *2023, 2024*
- Europlanet Summer School on Ground-based Observations and Science Communication *2023*
- Workshop on General Relativity, Cosmology and Astrophysics, IST

Teaching and Professional Experience

- **Researcher** *2025–Present*
  Deep Space Initiative, USA

  Conducting thermo-mechanical simulations in COMSOL and ANSYS to evaluate micrometeoroid and thermal impacts on space-habitat materials.
- **Physics Lecturer** *2024*
  Government Graduate College Chichawatni, Pakistan

  Delivered undergraduate lectures in mechanics, optics, and modern physics; supervised student laboratory projects.
- **Lab Associate** *2023–2024*
  Institute of Space Technology, Islamabad, Pakistan

  Taught observational astronomy and telescope operation courses to undergraduate Space Science students; supervised laboratory modules and practical observing sessions.

Honors and Awards

- Master's Research Grant, Institute of Space Technology *2022*
- Preliminary Asteroid Discovery, International Astronomical Search Collaboration (IASC) for identifiction a new asteroid candidate during a global student observing campaign *2021*
- Prime Minister's Youth Laptop Scheme, Government of Pakistan awarded to high-achieving university students on academic merit *2018*
- Merit Scholarship, Bahauddin Zakariya University *2015–2019*

Professional Affiliations

- Member, American Association of Variable Star Observers (AAVSO)

References

- **Dr. Shaukat Goderya**, Professor and Director, Programs for Astronomy Education and Research
  Tarleton State University (TSU), Texas, USA
  goderya@tarleton.edu